

Technical Project Report

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Low-light Image Enhancement via Zero-Reference Deep Curve Estimation (Zero-DCE)

1. Project Overview

Background and Motivation In many practical scenarios, image quality is severely compromised by poor lighting conditions, presenting a significant challenge for the extraction of meaningful information. Low-light environments typically result in images that are noisy, poorly illuminated, and devoid of essential detail. This degradation directly impacts the visibility and accuracy of downstream computer vision tasks. Enhancing images captured in these conditions is not merely a matter of aesthetics but is crucial for ensuring reliable image analysis in mission-critical applications.

Problem Statement The objective of this research is to develop a robust methodology for enhancing images captured in low-light environments. Effective enhancement is a prerequisite for diverse fields, including personal photography, surveillance, autonomous navigation, and medical imaging, where traditional processing methods—often relying on static or global adjustments—fail to yield satisfactory results. To overcome these limitations, we explore a deep learning-based approach that can intelligently restore clarity without the need for paired reference data.

Project Objectives

- **Develop a Deep Curve Estimation Network:** Design a deep learning architecture capable of predicting adaptive, image-specific enhancement curves.
- **Effectiveness Evaluation:** Rigorously evaluate the proposed method's ability to improve image quality under adverse lighting conditions.
- **Practical Application Exploration:** Investigate the utility of enhanced visual data in specialized sectors such as security and healthcare.
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2. Literature Review and Technical Context

Current State of Deep Learning for Enhancement Recent advancements in computer vision have seen a decisive shift from traditional image processing toward deep learning-based adaptive mappings. While traditional techniques struggle with the non-linear complexity and

variability of low-light scenes, deep learning allows models to learn sophisticated representations that intelligently enhance visibility and detail by understanding the intrinsic features of the input data.

The Shift to Curve Estimation Traditional deep learning models often attempt to learn direct image-to-image mappings. However, this project favors the shift toward curve estimation. By learning adaptive mappings rather than uniform adjustments, the network can adjust pixel values dynamically based on local context. This ensures that enhancements are localized and precise, preserving the original texture and color fidelity that global mappings often destroy.

3. Baseline Methodology: Basic CNN Approach

Initial Architecture Details Initial experimentation utilized a basic CNN encoder-decoder network to evaluate low-light feature extraction capabilities. This sequential model was constructed using a series of convolutional layers without skip-connections. The following Keras/TensorFlow implementation outlines the baseline structure:

```
model = Sequential([
    # encoder network
    Conv2D(256, 3, activation='relu', padding='same', input_shape=(256, 256, 3)),
    # MaxPooling2D(2, padding='same'),
    Conv2D(64, 3, activation='relu', padding='same'),
    # MaxPooling2D(2, padding='same'),
    # decoder network
    Conv2D(64, 3, activation='relu', padding='same'),
    # UpSampling2D(2),
    Conv2D(256, 3, activation='relu', padding='same'),
    # UpSampling2D(2),
    # output layer
    Conv2D(3, 3, activation='sigmoid', padding='same')
])

model.compile(optimizer='adam', loss='mean_squared_error')
model.summary()
```

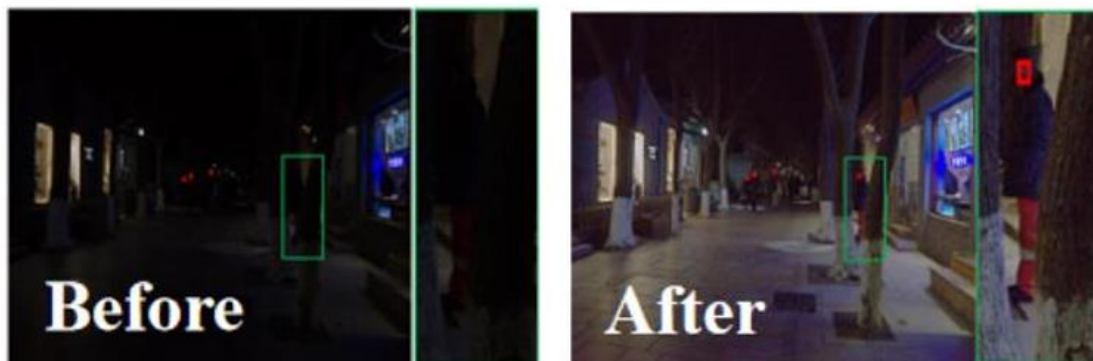
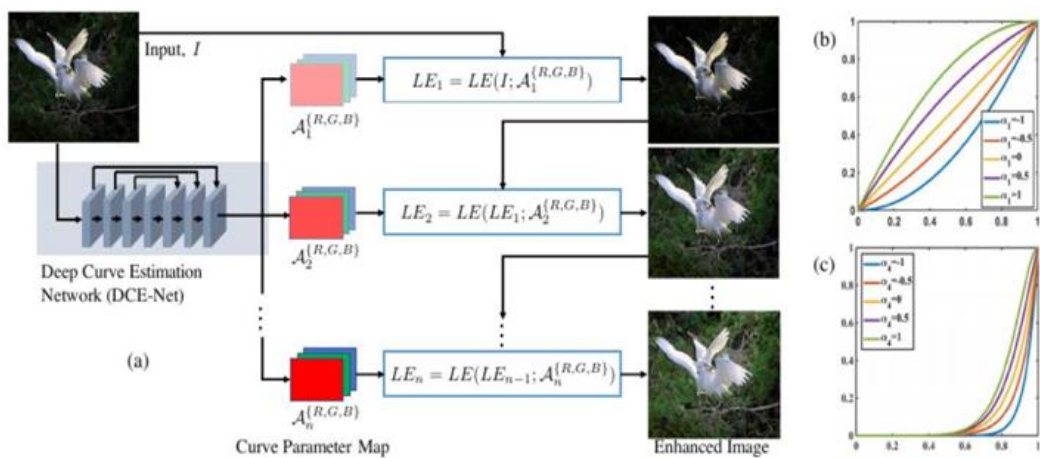
Performance Limitations The baseline architecture proved insufficient for the task, yielding the following results:

- **Suboptimal Metrics:** The model achieved a PSNR score of only **21.029**.
- **Technical Rationale for Failure:** The sequential nature of the encoder-decoder, specifically the lack of skip-connections, prevented the propagation of fine-grained spatial features to the deeper layers. Consequently, the network failed to capture the intricate structural details necessary for high-quality restoration. Removing Max Pooling layers in an attempt to retain spatial resolution provided negligible improvement, as the underlying architecture remained too simplistic to generalize without overfitting.

Data Augmentation and Transition To mitigate overfitting, we implemented a data augmentation strategy. We concatenated images along the RGB axis, increasing the input from 3 to 6 channels. This was a critical technical step to ensure absolute **spatial**

correspondence between the low-light and ground-truth pairs during random cropping or flipping. However, training on this augmented 6-channel dataset became unfeasible due to computational resource constraints. More importantly, this highlighted the fundamental limitation of reference-based training: the requirement for perfectly aligned high/low pairs is often impossible to satisfy in real-world deployments, necessitating a transition to the "Zero-Reference" paradigm.

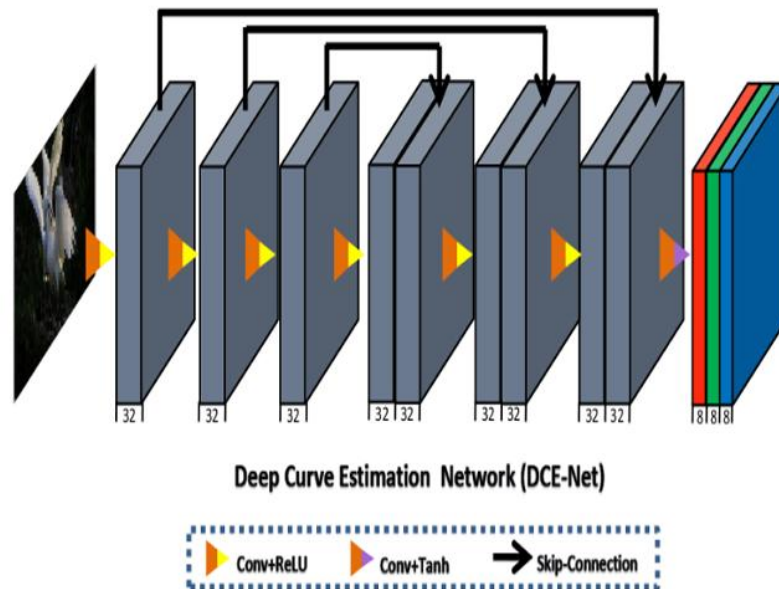
4. Proposed Framework: Zero-Reference Deep Curve Estimation (Zero-DCE)



Key Features of Zero-DCE

- **Image-Specific Curve Estimation:** The framework predicts high-order, pixel-wise curves tailored to the specific characteristics of each input image.
- **Zero-Reference Training:** Training is conducted without any paired or unpaired reference images, utilizing non-reference loss functions to evaluate quality.
- **Dynamic Range Adjustment:** The estimated curves map input pixels to a wider dynamic range, providing localized brightness and contrast improvements.

DCE-Net Architecture The DCE-Net is a lightweight network consisting of seven convolutional layers. It utilizes skip-connections to concatenate features from earlier layers, preserving spatial consistency. The network predicts the parameters for the enhancement curves rather than the image itself.



```
conv_layer1 = layers.Conv2D(
    filters=32, kernel_size=(3, 3), strides=(1, 1), activation="relu", padding="same",
)(input_layer)

conv_layer2 = layers.Conv2D(
    filters=32, kernel_size=(3, 3), strides=(1, 1), activation="relu", padding="same",
)(conv_layer1)

conv_layer3 = layers.Conv2D(
    filters=32, kernel_size=(3, 3), strides=(1, 1), activation="relu", padding="same",
)(conv_layer2)

conv_layer4 = layers.Conv2D(
    filters=32, kernel_size=(3, 3), strides=(1, 1), activation="relu", padding="same",
)(conv_layer3)

# Concatenation of the third and fourth convolutional layers
concat1 = layers.Concatenate(axis=-1)([conv_layer4, conv_layer3])

# Fifth convolutional layer after concatenation
conv_layer5 = layers.Conv2D(
    filters=32, kernel_size=(3, 3), strides=(1, 1), activation="relu", padding="same",
)(concat1)

# Concatenation of the fifth layer with the second convolutional layer
concat2 = layers.Concatenate(axis=-1)([conv_layer5, conv_layer2])

# Sixth convolutional layer after the second concatenation
conv_layer6 = layers.Conv2D(
    filters=32, kernel_size=(3, 3), strides=(1, 1), activation="relu", padding="same",
)(concat2)
```

Light-Enhancement (LE) Curve Dynamics The framework relies on a fundamental quadratic curve:where x is the normalized input pixel value and y is the predicted adjustment parameter. By applying this curve iteratively (N iterations), the network effectively approximates a **high-order polynomial mapping**. This iterative process allows for more complex, non-linear transformations that provide a gradual and controlled increase in dynamic range.

5. Non-Reference Loss Functions

Zero-DCE enables training without ground-truth images by employing four non-reference loss functions based on perceptual and structural constraints.

1. **Exposure Control Loss ($\mathcal{L}_E$):** Measures the deviation of local region average intensity from a desired well-exposed level μ_0 . The loss is computed over non-overlapping local regions of size k :
2. **Color Preservation Loss ($\mathcal{L}_C$):** Based on the Gray-World constancy hypothesis, this loss ensures color fidelity by minimizing differences between average channel intensities. We define the set of channel pairs $\mathcal{P}$.
3. **Spatial Consistency Loss ($\mathcal{L}_S$):** Preserves the structural integrity by maintaining the difference in intensity between a pixel and its neighbors across k local regions in both the original (I) and enhanced ($\hat{I}$) images:
4. **Illumination Smoothness Loss ($\mathcal{L}_I$):** To ensure natural-looking results, this loss is applied to the **curve parameter map ($\mathbf{p}$)** to penalize abrupt illumination changes and maintain monotonicity.

6. Evaluation and Results

Comparative Analysis The shift from the sequential CNN to the iterative DCE-Net provided a marked increase in performance and structural awareness.

Feature	Initial CNN Model	Zero-DCE Model
Architecture	Sequential (No Skips)	7-Layer Conv (With Skips)
Complexity	Simple Encoder-Decoder	Lightweight/Iterative
PSNR Score	21.029	27.602611571511055

Visual Evidence The Zero-DCE model demonstrates superior ability in lifting shadows and revealing hidden objects while effectively managing noise. Unlike the baseline, which often produced washed-out or artifact-heavy results, Zero-DCE preserves the natural color balance and texture, confirming the efficacy of iterative curve estimation.



Final Performance Metric The primary success of the project is quantified by the achieved PSNR score of **27.602611571511055**, validating the Zero-Reference approach as a scalable and efficient solution for real-world low-light enhancement.

```
def calculate_psnr(original_image, enhanced_image):
    """
    Calculate the Peak Signal-to-Noise Ratio (PSNR) between two images.

    Parameters:
    - original_image (np.ndarray): The original reference image.
    - enhanced_image (np.ndarray): The enhanced image to compare.

    Returns:
    - psnr (float): The PSNR value in decibels.
    """
    # Ensure the input images are in the correct format and dimensions
    original_image = np.array(original_image, dtype=np.float32)
    enhanced_image = np.array(enhanced_image, dtype=np.float32)

    if original_image.shape != enhanced_image.shape:
        raise ValueError("Input images must have the same dimensions and channels")

    # Compute the Mean Squared Error (MSE)
    mse = np.mean((original_image - enhanced_image) ** 2)

    if mse == 0:
        return float('inf') # If the MSE is zero, the PSNR is infinite

    # Set the maximum pixel value
    max_pixel_value = 255.0

    # Compute the PSNR
    psnr = 10 * np.log10(max_pixel_value / np.sqrt(mse))

    return psnr
```

7. References

Primary Research

- Zero-DCE Project and Repository: https://li-chongyi.github.io/Proj_Zero-DCE.html

Secondary References

- Zhang et al., "Deep Color Consistent Network for Low-Light Image Enhancement," CVPR 2022. [Link](#)
- Implementation Tutorial: [DCE-Net Guide](#)